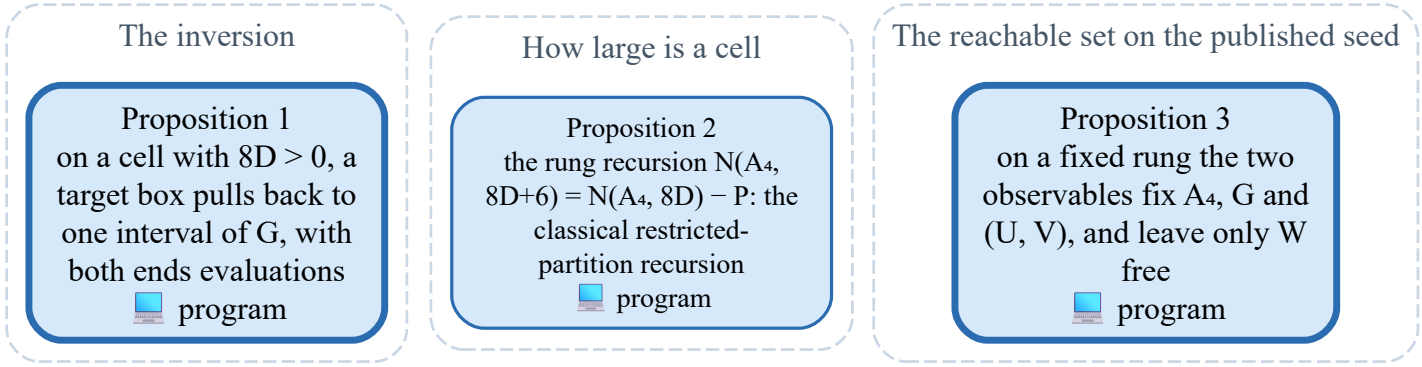
🗺️ Map of *A Certified 2.68 TeV Gap in the Closed-Form Map of the Compactification Scale in SU(7) Grand Gauge-Higgs Unification, and the Invariant Its Two Observables Cannot See (Part VIII)*

🏷️ **High-energy physics · grand gauge-Higgs unification**

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3 results | ✅ **0** formalized in Lean | ◐ **0** with a part in Lean | 💻 **3** checked by a program | written proof only | (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



What the paper contributes

- **The inverse map is decidable** on the eight-generator lattice of bulk contents of the SU(7) model of Komori and Maru: a target box in $(1/R_5,m_h)$ pulls back to finitely many cells and, on each, one interval of the moment G (Proposition 1); a cell is finite because only $\mathbf 7(+,+)$ has $A_4=0$ (Observation 1 of the paper).
- **"No content" carries a named certificate**: cone, congruence, an exact rational Farkas pair, exhaustion, or *floor*. One dual certificate is printed in full.
- **A certified gap of the closed-form map**: on the published seed the enumerated image is 35 points in $[8897.5,9967]$ GeV ($8D=1$) and 65 in $[5258,6215]$ ($8D=3$), and nothing lies in $(6.216,\,8.897)$ TeV; at the measured Higgs mass one point, $1/R_5=8967.5$ GeV, carried by 81 contents with the same one-loop potential.
- **What the observables cannot see**: on a fixed rung $(1/R_5,m_h)$ fix four of the five complete invariants (Proposition 3), but two contents share both and disagree about the vacuum invariant W.
- Scope stated by the paper: every emptiness is a statement about the closed form, not yet the untruncated model; the recursion of Proposition 2 is classical (Beck–Robins).

🚀 **Start here: Proposition 1**: on a cell with 8D > 0, a target box pulls back to one interval of G, with both ends evaluations.

Main results:

- **Proposition 1**: on a cell with 8D > 0, a target box pulls back to one interval of G, with both ends evaluations
- **Proposition 3**: on a fixed rung the two observables fix A_4, G and (U, V), and leave only W free

🧭 Fields: where the paper lives and what it connects

Tools taken from other branches

- **Integer and linear programming** → exact rational Farkas certificates over the relaxation of a cell; the moment cone and the mod-6 congruence (Schrijver) — Proposition 1
- **Vector partition functions** → counting a cell without building it, by dynamic programming; the classical restricted-partition recursion (Sturmfels; Beck–Robins) — Proposition 2
- **Elementary number theory** → unique factorisation: 2^a 3^b = 1 forces a = b = 0 — Proposition 3

What the results serve

- **Phenomenology of gauge-Higgs unification** ← the certified gap (6.216, 8.897) TeV of the closed-form map, a single $1/R_5 = 8967.5$ GeV at the measured Higgs mass, and its image in the CMS dijet angular $\Delta\chi^2$ — Proposition 1
- **Inverse problems from observables to model space** ← a decidable inverse map with certificates, and an invariant (W) the two fitted observables cannot see — Proposition 3